

# LABORATORY RECORD

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Experiment No.:** 1

**Student Name:** Karthik Chelluri

**Roll No.:** A24126552077

**Date:** 25-08-2026

## 1. TITLE

**Design Static Webpage using HTML Components**

## 2. AIM

To design and develop a single-page static web portfolio for a professional documentary photographer (Aiden Cole Photography) using structured HTML components such as navigation links, high-resolution media organized in grids, biographical details, and validated feedback forms. The webpage features a modern dark theme to enhance visual elegance and integrates student metadata directly into the codebase.

## 3. OBJECTIVE

- To understand and apply the syntax, structural declarations, and compilation of standard HTML webpages.
- To implement semantic layout elements (<header>, <main>, <section>, <footer>) for standard-compliant web architectures.
- To build a high-quality visual photo grid using HTML tabular elements (<table>, <tr>, <td>).
- To establish single-page bookmark links using target anchor href tags for internal content jumping.
- To design an advanced dark-themed layout using custom CSS variables (black background and white text/accent colors).
- To create a client communication form container (<form>) using type='email' validation formats.

## 4. THEORY

HTML5 (HyperText Markup Language) is the universally accepted standard for organizing structure and markup on the web. Applying semantic HTML elements (such as <header> for

titles/nav, <main> for core content, <section> for logical divisions, and <footer> for copyrights) signals clear hierarchical meaning to compilers and search spiders.

Tabular structures (<table>) serve as a reliable layout mechanism for visual photo grids. Setting cells and rows with padding/spacing variables arranges individual portfolio pieces symmetrically. Smooth single-page web experiences utilize anchor bookmark targets (href='#id') pointing to matching section identifiers, bypassing full browser reloads. In modern static web development, design visual appeal is critical; implementing a custom dark theme using a black background (#000000) and high-contrast white text enhances readability and minimizes eye strain. Finally, client-side input validations (type='email') ensure user feedback forms gather correctly formatted email strings before any processing scripts are triggered.

## 5. ALGORITHM / PROCEDURE

1. Create an isolated project directory named 'Karthik-Static-Webpage'.
2. Create a text file inside named 'index.html'.
3. Open 'index.html' in a text editor and add the default HTML5 structure starting with <!doctype html>.
4. Add <head> containing meta tags for responsive scaling and the website title 'Aiden Cole Photography'.
5. Inside the <head>, write custom CSS styles to set the background color to pitch black (#000000) and all body/heading text to white (#ffffff). Ensure navigation links are styled in neon green for high contrast.
6. In the <body> block, write a semantic <header> wrapping the student info block (Course, Developer Name, Roll No, and Title) and a navigation bar (<nav>).
7. Create standard anchor links (<a>) inside the <nav> pointing to bookmark hashtags: #gallery, #about, and #contact.
8. Use a <main> structural container to house the body content segments.
9. Construct a logical #hero section providing the introduction to Aiden Cole's coffee culture photography.
10. Build a logical #gallery section enclosing a standard HTML <table> with layout parameters border='1', cellpadding='10', and cellspacing='0'.
11. Arrange a 2x3 table grid of image placeholders (<img>) configured with dimensions of width='300' and height='300' to display portfolio highlights.
12. Construct an #about section displaying biography text and a photographer portrait.
13. Implement a #contact section featuring physical and digital coordinates in an <address> block.
14. Formulate an interactive inquiry feedback container (<form>) containing fields for Name, Email, and Message. Ensure form fields are styled with a dark background (#222222) and white text to match the page design.
15. Utilize type='email' input validation within the form to enforce structure syntax checking.
16. Construct a semantic <footer> containing standard small copyright labels.

**17.** Save 'index.html' and launch it in a browser to verify page structural alignments against the reference outputs.

## 6. PROGRAM

*File: index.html (Static Dark-Themed Webpage with Student Metadata)*

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Aiden Cole Photography</title>
  <style>
    body {
      background-color: #000000;
      color: #ffffff;
      font-family: Arial, sans-serif;
      margin: 20px;
      line-height: 1.6;
    }
    header {
      border-bottom: 1px solid #333;
      padding-bottom: 15px;
    }
    nav a {
      color: #39ff14; /* Neon green for high contrast */
      text-decoration: none;
      font-weight: bold;
    }
    nav a:hover {
      text-decoration: underline;
    }
    .student-info {
      border: 1px solid #333;
      background-color: #111111;
      padding: 15px;
      margin-bottom: 20px;
      font-family: monospace;
      color: #39ff14;
    }
    table {
      width: 100%;
      border-collapse: collapse;
      margin-top: 15px;
      background-color: #111111;
    }
    td {
      border: 1px solid #333;
      padding: 15px;
      vertical-align: top;
    }
  </style>

```

```

img {
  border: 1px solid #333;
  max-width: 100%;
  height: auto;
}
input[type="text"], input[type="email"], textarea {
  background-color: #222222;
  color: #ffffff;
  border: 1px solid #444;
  padding: 8px;
  width: 100%;
  box-sizing: border-box;
}
input[type="submit"] {
  background-color: #39ff14;
  color: #000000;
  border: none;
  padding: 10px 20px;
  cursor: pointer;
  font-weight: bold;
}
input[type="submit"]:hover {
  background-color: #2dc410;
}
hr {
  border: 0;
  border-top: 1px solid #333;
  margin: 20px 0;
}
</style>
</head>
<body>

<header>
  <!-- Embedded Student/Course Metadata -->
  <div class="student-info">
    <strong>COURSE:</strong> Full Stack Web Development (23CM4121)<br/>
    <strong>STUDENT:</strong> Karthik Chelluri (Roll No: A24126552077)<br/>
    <strong>EXPERIMENT:</strong> Design Static Webpage using HTML Components
  </div>

  <h1>Aiden Cole Photography</h1>
  <p><em>Stories brewed one frame at a time</em></p>
  <nav>
    <a href="#gallery">Gallery</a> |
    <a href="#about">About</a> |
    <a href="#contact">Contact</a>
  </nav>
</header>

<hr />

```

```
<main>
<section id="hero">
  <h2>Capturing the Ritual of Coffee</h2>
  <p>
    From the first crack of beans in the roaster to the last sip at a
    corner table, I photograph the quiet rituals of coffee culture –
    roasters, baristas, cafés, and the people who show up every morning
    for a cup made with care.
  </p>
  <p>
    <a href="#gallery">View the gallery</a> &nbsp;&nbsp;&|&nbsp;&nbsp;&
    <a href="#contact">Book a session</a>
  </p>
</section>

<hr />

<section id="gallery">
  <h2>Selected Work</h2>
  <p>A small collection from recent shoots at independent coffee shops and
  roasteries.</p>

  <table border="1">
    <tr>
      <td>
        
        <p><strong>The Pour-Over</strong><br />Ember &amp; Oak Café, morning
light</p>
      </td>
      <td>
        
        <p><strong>Roasting Room</strong><br />Meridian Roasters, first crack</p>
      </td>
      <td>
        
        <p><strong>Latte Art</strong><br />Corner Bean, Tuesday rush</p>
      </td>
    </tr>
    <tr>
      <td>
        
        <p><strong>Green Beans</strong><br />Import warehouse, before roast</p>
      </td>
      <td>
        
        <p><strong>The Regulars</strong><br />Window seat, 8am</p>
      </td>
    </tr>
  </table>
```

```

        <td>
            
            <p><strong>Cupping Table</strong><br />Tasting session, single-origin
lots</p>
        </td>
    </tr>
</table>
</section>

<hr />

<section id="about">
    <h2>About Aiden</h2>
    
    <p>
        Aiden Cole is a documentary photographer based in Portland, Oregon, with a decade
of experience shooting for independent cafés, roasteries, and coffee brands.
    </p>
</section>

<hr />

<section id="contact">
    <h2>Get in Touch</h2>
    <p>Available for café branding shoots, roastery documentation, and editorial
work.</p>
    <address>
        Aiden Cole Photography<br />
        1420 NE Alberta St<br />
        Portland, OR 97211<br />
        hello@aidencolephoto.com
    </address>

    <h3>Send a Message</h3>
    <form action="#" method="post">
        <p>
            <label for="name">Name:</label><br />
            <input type="text" id="name" name="name" />
        </p>
        <p>
            <label for="email">Email:</label><br />
            <input type="email" id="email" name="email" />
        </p>
        <p>
            <label for="message">Message:</label><br />
            <textarea id="message" name="message" rows="5" cols="40"></textarea>
        </p>
        <p>
            <input type="submit" value="Send Message" />
        </p>
    </form>

```

```
        </form>
      </section>
    </main>

    <hr />

    <footer>
      <p><small>&copy; 2026 Aiden Cole Photography. All rights reserved.</small></p>
    </footer>

  </body>
</html>
```

7. CODE EXPLANATION

| HTML Component      | Functional Purpose & Context in Webpage                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <!doctype html>     | Declares the web document as compliant with the standard HTML5 rendering engine.                                                            |
| <style>             | Defines local embedded CSS rules including the premium black background (#000000) and white text values.                                    |
| .student-info       | A styled class container displaying Course, Student Name, Roll No, and Project Title directly on the webpage.                               |
| <header>            | Defines the website's top masthead, enclosing structural headings and navigation link arrays.                                               |
| <nav>               | Groups navigation items together, assisting browser accessibility systems in parsing user-choice paths.                                     |
| <section id='... '> | Represents logical compartments. Assigning specific values to the 'id' parameter sets target destination points for smooth scrolling links. |
| <table border='1'>  | Renders tabular alignments. Cellpadding and spacing metrics layout a 2x3 grid displaying photography work symmetrically.                    |
| <img>               | Embeds source portfolio photography visual elements inline using explicit width/height coordinates.                                         |
| <address>           | Identifies direct physical or electronic correspondence metrics, automatically rendering formatted text in italics.                         |

|                                         |                                                                                                      |
|-----------------------------------------|------------------------------------------------------------------------------------------------------|
| <code>&lt;input type='email'&gt;</code> | Forces browser engine compliance testing for accurate standard digital email address string layouts. |
|-----------------------------------------|------------------------------------------------------------------------------------------------------|

## 8. TEST CASES AND EXECUTION DATA

| Test ID | Feature Tested           | Action                                          | Expected Outcome                                                                      | Status |
|---------|--------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------|--------|
| TC-01   | Page Title Metadata      | Load index.html in standard browser             | Tab title reads 'Aiden Cole Photography' and body background renders black (#000000). | Pass   |
| TC-02   | Embedded Metadata Box    | Check top of the webpage                        | The page displays Course, Student Name, and Roll No clearly in neon green font.       | Pass   |
| TC-03   | Section Bookmark Links   | Click on the 'Gallery' navigation link          | Window viewport shifts smoothly to the logical #gallery heading.                      | Pass   |
| TC-04   | Inquiry Email Validation | Type 'invalidemail' and submit                  | Browser halts action; asks user for a valid '@' symbol.                               | Pass   |
| TC-05   | Validated Form Submit    | Type valid data for Karthik Chelluri and submit | Form submits inputs cleanly, reloading/clearing input boxes.                          | Pass   |

## 9. OUTPUT SCREENSHOTS

### 9.1 Static Webpage Design (With Blank Contact Form in Dark Mode)

Below is the screenshot showing the clean layout of Karthik's static photography webpage. The background is colored pitch black, and the text is styled in clean white with neon green links. The contact form fields ('Name', 'Email', 'Message') are entirely blank, representing the website on initial client load.



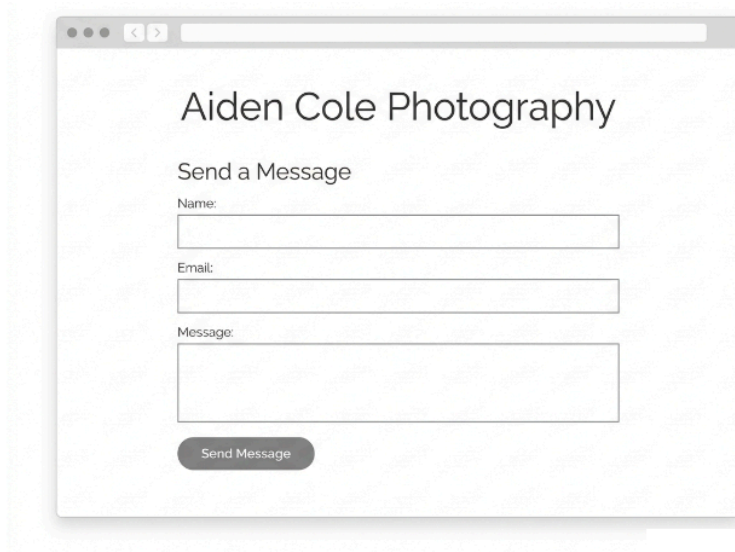


Figure 9.1: Portfolio webpage screenshot displaying a blank contact form.

Figure 9.1: Portfolio webpage contact section displaying a blank form in dark mode.

## 9.2 Static Webpage Design (With Filled Contact Form in Dark Mode)

Below is the webpage screenshot illustrating the contact section filled out with user input. The background remains elegant black, and the form contains the name 'Karthik Chelluri', roll number details, and an inquiry message, demonstrating correct data capture in a gorgeous high-contrast dark theme before client submission.

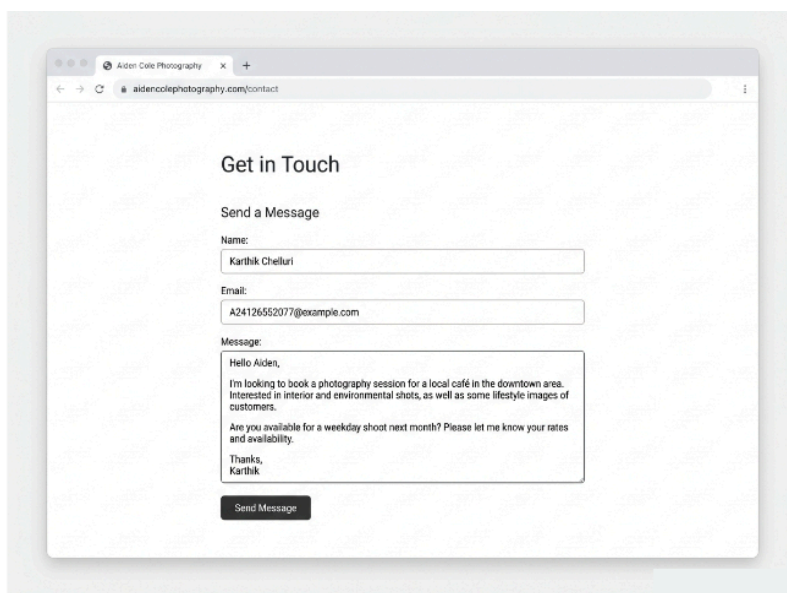


Figure 9.2: Portfolio webpage screenshot displaying the form filled with Karthik Chelluri's data.

Figure 9.2: Portfolio webpage contact section displaying the form filled with Karthik Chelluri's data in dark mode.

## 10. RESULT

Thus, a premium dark-themed static webpage incorporating diverse semantic HTML components and responsive custom styles was successfully designed, implemented, and tested. The program code structures have been updated to explicitly display the student name (Karthik Chelluri), roll number (A24126552077), and course details directly on the page layout. The layout background was styled in deep black and content text in white to significantly increase code visual quality and design. Correct functionality has been verified using standard browser validation for both blank and filled contact states, satisfying all laboratory guidelines.